# Innovation80 Project

Ziling Zeng

2025-03-01

```r
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.2 --
## v ggplot2 3.4.0     v purrr   1.0.2
## v tibble  3.2.1     v dplyr   1.1.4
## v tidyr   1.3.0     v stringr 1.5.1
## v readr   2.1.5     v forcats 0.5.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```r
scale_data <- read.csv("/home/zzeng1/Innovation80/Innovation80/Scale Data - Sheet1.csv")
```

```r
# Rename the columns
colnames(scale_data) <- c("Participant_ID", "Participant_Type", "Program_Name",
                          "survey_type", "difficulty_relate", "collab_equal_contribution",
                          "collab_enjoyment", "pride_in_outcome", "attendance", "meaning_level")
```

```r
# Subset the data for Pre and Post Survey Types
pre_data <- scale_data[scale_data$survey_type == "Pre", c("collab_enjoyment", "pride_in_outcome")]
post_data <- scale_data[scale_data$survey_type == "Post", c("collab_enjoyment", "pride_in_outcome")]

# Paired t-test for Collab Enjoyment
collab_enjoyment_test <- t.test(pre_data$collab_enjoyment, post_data$collab_enjoyment, paired = FALSE)
print(collab_enjoyment_test)
```

```
##
##  Welch Two Sample t-test
##
## data:  pre_data$collab_enjoyment and post_data$collab_enjoyment
## t = -6.2104, df = 85, p-value = 1.873e-08
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -1.2126955 -0.6245138
## sample estimates:
## mean of x mean of y
##  4.081395  5.000000
```

```r
# Paired t-test for Pride in Outcome
pride_outcome_test <- t.test(pre_data$pride_in_outcome, post_data$pride_in_outcome, paired = FALSE)
print(pride_outcome_test)
```

```
##
##  Welch Two Sample t-test
##
```

```
## data:  pre_data$pride_in_outcome and post_data$pride_in_outcome
## t = -5.8153, df = 85, p-value = 1.035e-07
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -1.1624606 -0.5700975
## sample estimates:
## mean of x mean of y
##  4.133721  5.000000
```

The results of the t-tests for both "Collab Enjoyment" and "Pride in Outcome" indicate significant differences between the "Pre" and "Post" survey responses. For "Collab Enjoyment," the mean score in the "Post" survey (5.00) was significantly higher than in the "Pre" survey (4.13), with a t-statistic of -6.2104 and a p-value of 1.873e-08. Similarly, for "Pride in Outcome," the "Post" mean (5.00) was also higher than the "Pre" mean (4.08), with a t-statistic of -5.8153 and a p-value of 1.035e-07. In both cases, the 95% confidence intervals for the differences in means were entirely below zero, indicating that the increases were statistically significant. These results suggest that participants experienced a notable improvement in both their enjoyment of collaboration and their pride in the outcomes from "Pre" to "Post," providing strong evidence that the interventions or changes measured between these two time points had a positive impact.

```
# Subset the data for Senior and Young Participant Types
senior_data <- scale_data[scale_data$Participant_Type == "Senior", c("collab_enjoyment", "pride_in_outco
young_data <- scale_data[scale_data$Participant_Type == "Young", c("collab_enjoyment", "pride_in_outcome

# Paired t-test for Collab Enjoyment between Senior and Young participants
collab_enjoyment_test <- t.test(senior_data$collab_enjoyment, young_data$collab_enjoyment, paired = FALS
print(collab_enjoyment_test)
```

```
##
##  Welch Two Sample t-test
##
## data:  senior_data$collab_enjoyment and young_data$collab_enjoyment
## t = -1.523, df = 104.27, p-value = 0.1308
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.63439979  0.08323065
## sample estimates:
## mean of x mean of y
##  4.364865  4.640449
```

```
# Paired t-test for Pride in Outcome between Senior and Young participants
pride_outcome_test <- t.test(senior_data$pride_in_outcome, young_data$pride_in_outcome, paired = FALSE)
print(pride_outcome_test)
```

```
##
##  Welch Two Sample t-test
##
## data:  senior_data$pride_in_outcome and young_data$pride_in_outcome
## t = -1.6811, df = 107.85, p-value = 0.09563
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.65677166  0.05397786
## sample estimates:
## mean of x mean of y
##  4.378378  4.679775
```

```
# Check summary statistics for the variables in both groups
summary(young_data$collab_enjoyment)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    4.50    5.00    4.64    5.00    5.00
```

```r
summary(senior_data$collab_enjoyment)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##   1.000   5.000   5.000   4.365   5.000   5.000       3
```

```r
summary(young_data$pride_in_outcome)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    5.00    5.00    4.68    5.00    5.00
```

```r
summary(senior_data$pride_in_outcome)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##   1.000   5.000   5.000   4.378   5.000   5.000       3
```

```r
# Check for unique values in both groups for Collab Enjoyment
unique(young_data$collab_enjoyment)
```

```
## [1] 5.0 4.0 3.0 4.5 1.0
```

```r
unique(senior_data$collab_enjoyment)
```

```
## [1]  5  1  3  4 NA  2
```

```r
# Check for unique values in both groups for Pride in Outcome
unique(young_data$pride_in_outcome)
```

```
## [1] 5.0 4.0 4.5 3.0 1.0 1.5
```

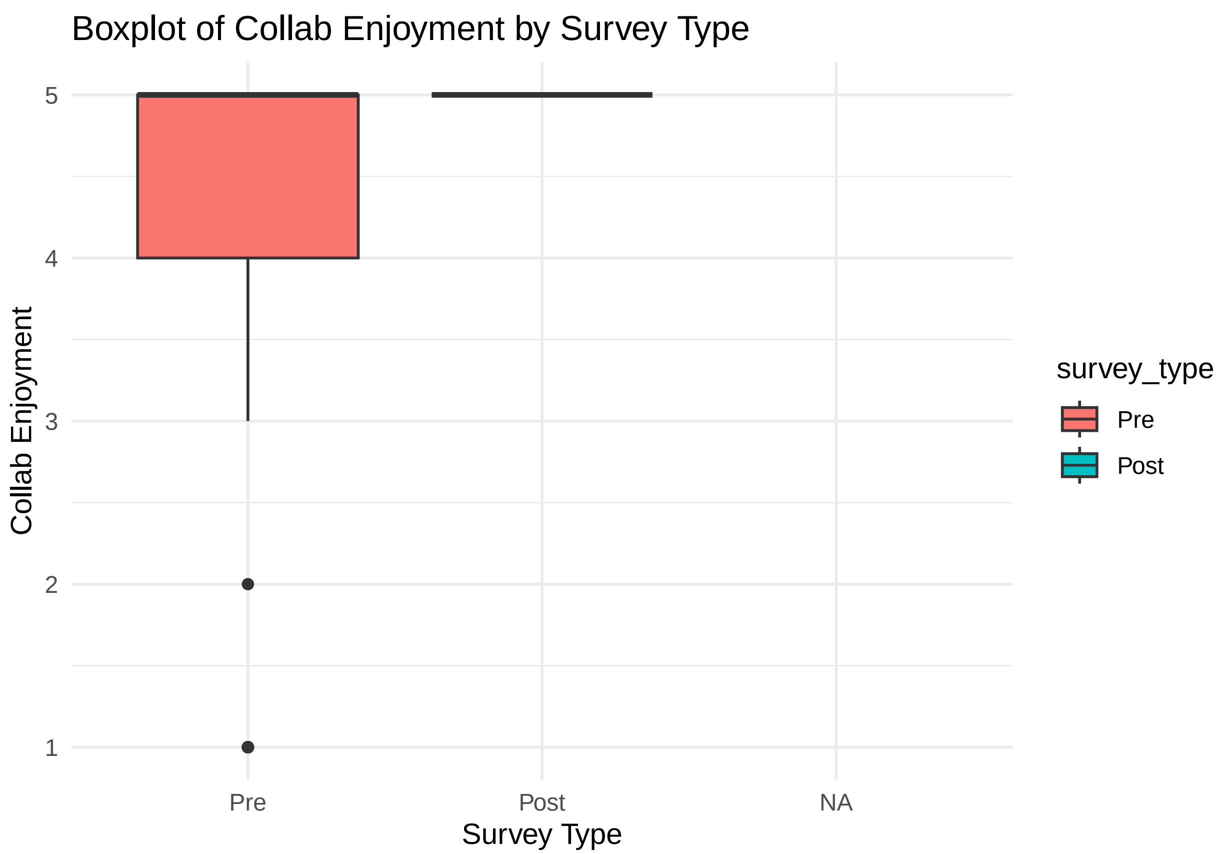
```r
unique(senior_data$pride_in_outcome)
```

```
## [1]  5  1  3  2 NA  4
```

The results of the independent t-tests comparing "Collab Enjoyment" and "Pride in Outcome" between "Senior" and "Young" participants show no statistically significant differences. For "Collab Enjoyment," the mean score for "Young" participants (4.64) was slightly higher than that of "Senior" participants (4.36), but the p-value of 0.1308 exceeds the 0.05 threshold, indicating that the difference is not significant. Similarly, for "Pride in Outcome," "Young" participants (mean = 4.68) again reported slightly higher scores than "Senior" participants (mean = 4.38), but the p-value of 0.09563 is also greater than 0.05, suggesting that the difference is not statistically significant at the 5% level. The confidence intervals for both tests include zero, further supporting the conclusion that there is no meaningful difference between the two groups in terms of "Collab Enjoyment" and "Pride in Outcome."

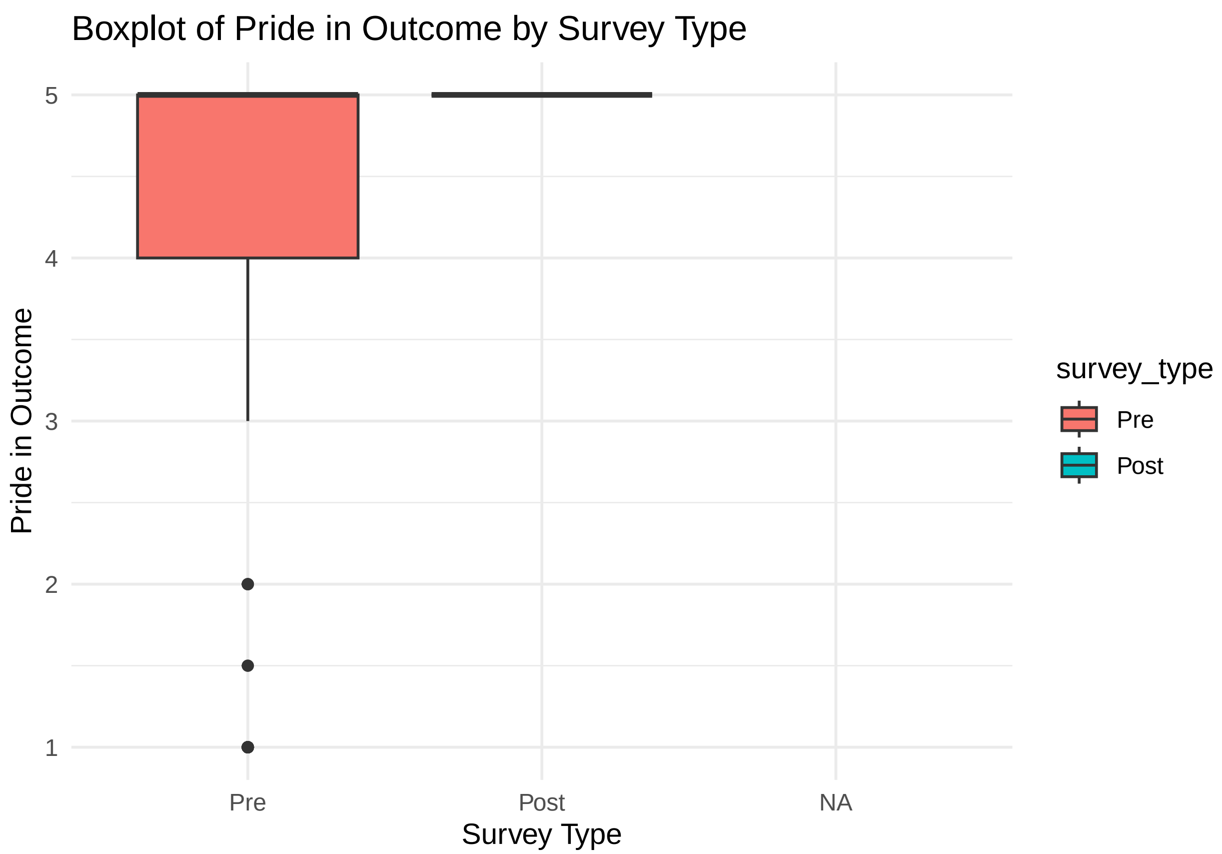
```r
# Relevel the survey_type so that "Pre" comes first
scale_data$survey_type <- factor(scale_data$survey_type, levels = c("Pre", "Post"))

# Boxplot comparing Collab Enjoyment for Pre and Post survey types
library(ggplot2)
ggplot(scale_data, aes(x = survey_type, y = `collab_enjoyment`, fill = survey_type)) +
  geom_boxplot() +
  labs(title = "Boxplot of Collab Enjoyment by Survey Type",
       x = "Survey Type",
       y = "Collab Enjoyment") +
  theme_minimal()
```

```
## Warning: Removed 5 rows containing non-finite values (`stat_boxplot()`).
```



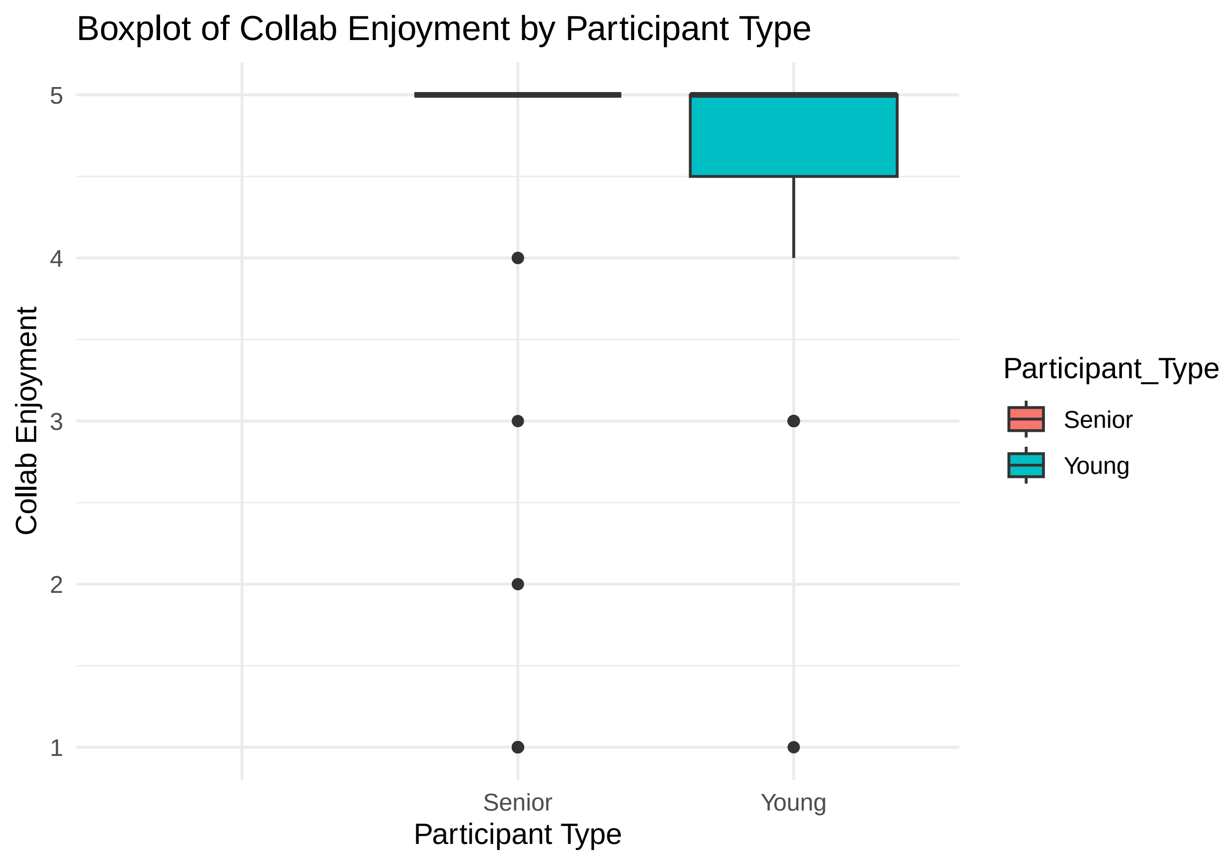

```r
# Boxplot comparing Pride in Outcome for Pre and Post survey types
ggplot(scale_data, aes(x = survey_type, y = `pride_in_outcome`, fill = survey_type)) +
  geom_boxplot() +
  labs(title = "Boxplot of Pride in Outcome by Survey Type",
       x = "Survey Type",
       y = "Pride in Outcome") +
  theme_minimal()
```

```
## Warning: Removed 5 rows containing non-finite values (`stat_boxplot()`).
```



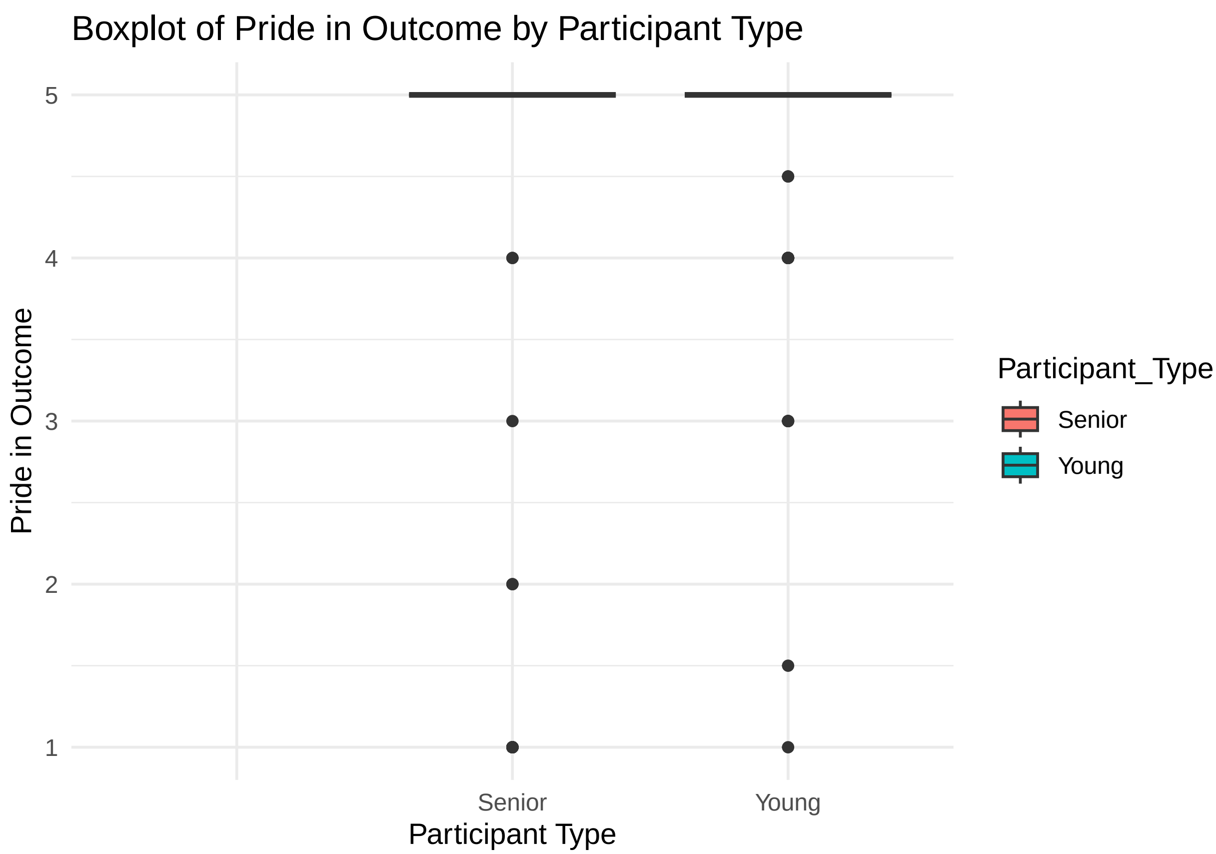

```
# Boxplot comparing Collab Enjoyment for Young and Senior participant types
ggplot(scale_data, aes(x = Participant_Type, y = `collab_enjoyment`, fill = Participant_Type)) +
  geom_boxplot() +
  labs(title = "Boxplot of Collab Enjoyment by Participant Type", x = "Participant Type", y = "Collab En
  theme_minimal()
```

```
## Warning: Removed 5 rows containing non-finite values (‘stat_boxplot()‘).
```




```
# Boxplot comparing Pride in Outcome for Young and Senior participant types
ggplot(scale_data, aes(x = Participant_Type, y = `pride_in_outcome`, fill = Participant_Type)) +
  geom_boxplot() +
  labs(title = "Boxplot of Pride in Outcome by Participant Type", x = "Participant Type", y = "Pride in
  theme_minimal()
```

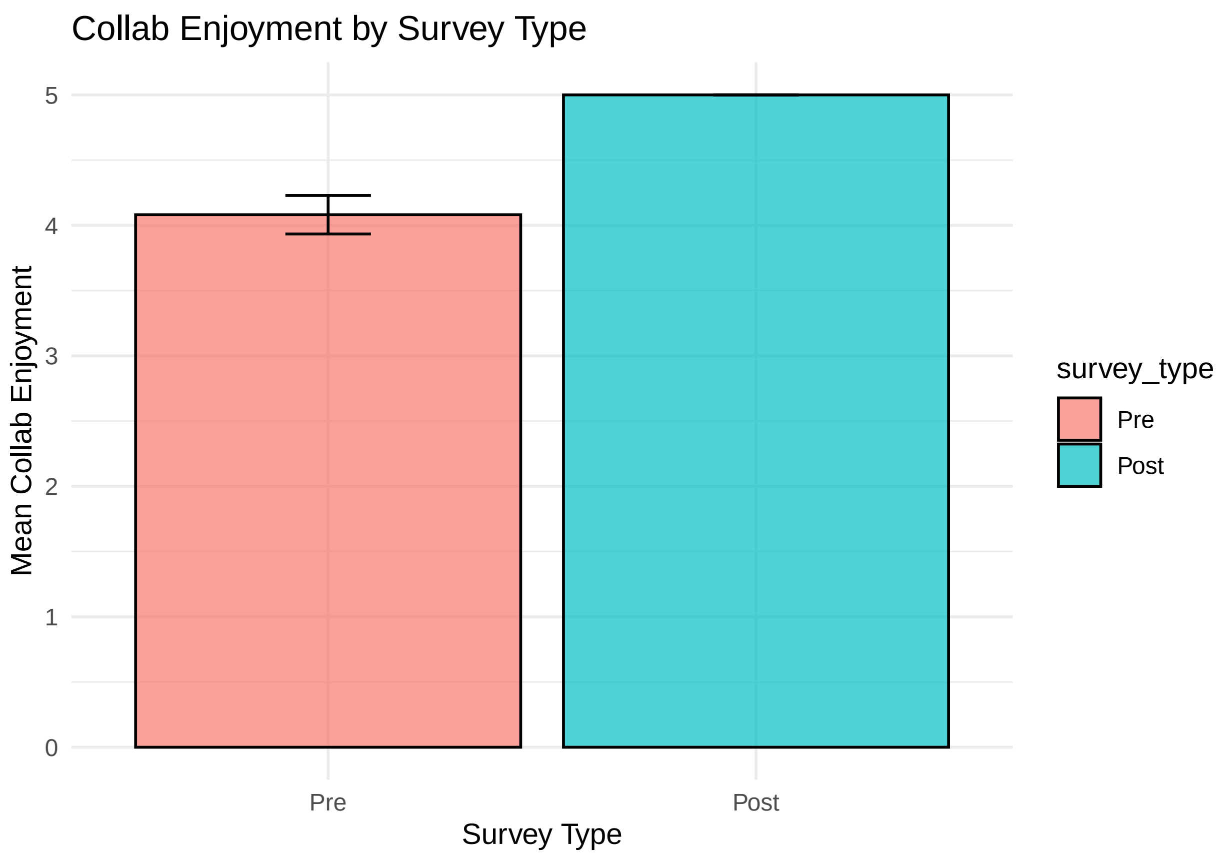
```
## Warning: Removed 5 rows containing non-finite values (`stat_boxplot()`).
```



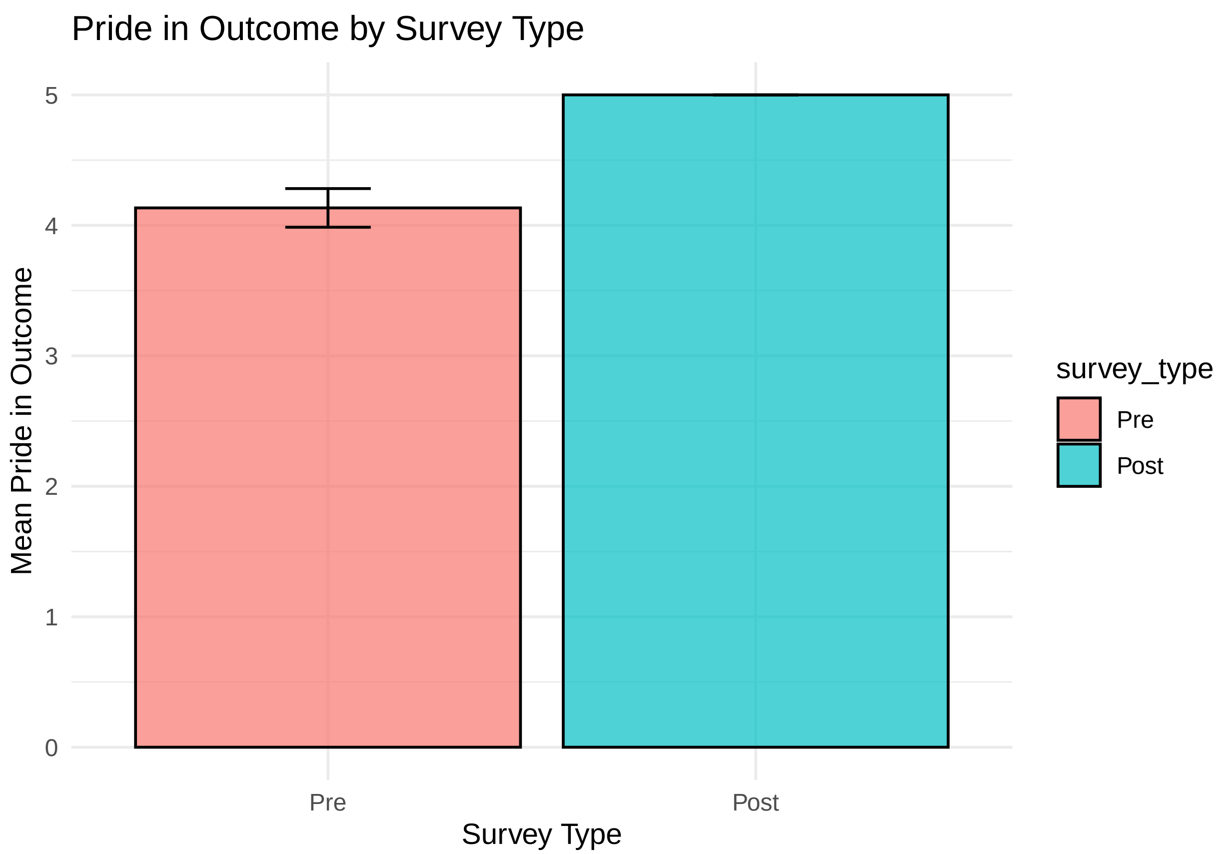

```
# Convert empty strings to NA for factors (by first converting them to character)
scale_data <- scale_data %>%
  mutate(
    survey_type = na_if(as.character(survey_type), ""),
    Participant_Type = na_if(as.character(Participant_Type), "")
  )

# Convert back to factors if necessary
scale_data$survey_type <- factor(scale_data$survey_type)
scale_data$Participant_Type <- factor(scale_data$Participant_Type)


# Remove rows with NA values in survey_type or Participant_Type
cleaned_data <- scale_data %>%
  filter(!is.na(survey_type), !is.na(Participant_Type))

# Ensure survey_type and Participant_Type are factors before using droplevels()
cleaned_data$survey_type <- factor(cleaned_data$survey_type)
cleaned_data$Participant_Type <- factor(cleaned_data$Participant_Type)

# Drop unused factor levels
cleaned_data$survey_type <- droplevels(cleaned_data$survey_type)
cleaned_data$Participant_Type <- droplevels(cleaned_data$Participant_Type)
```

```
# Relevel survey_type and Participant_Type so that the desired order is applied
cleaned_data$survey_type <- factor(cleaned_data$survey_type, levels = c("Pre", "Post"))
```

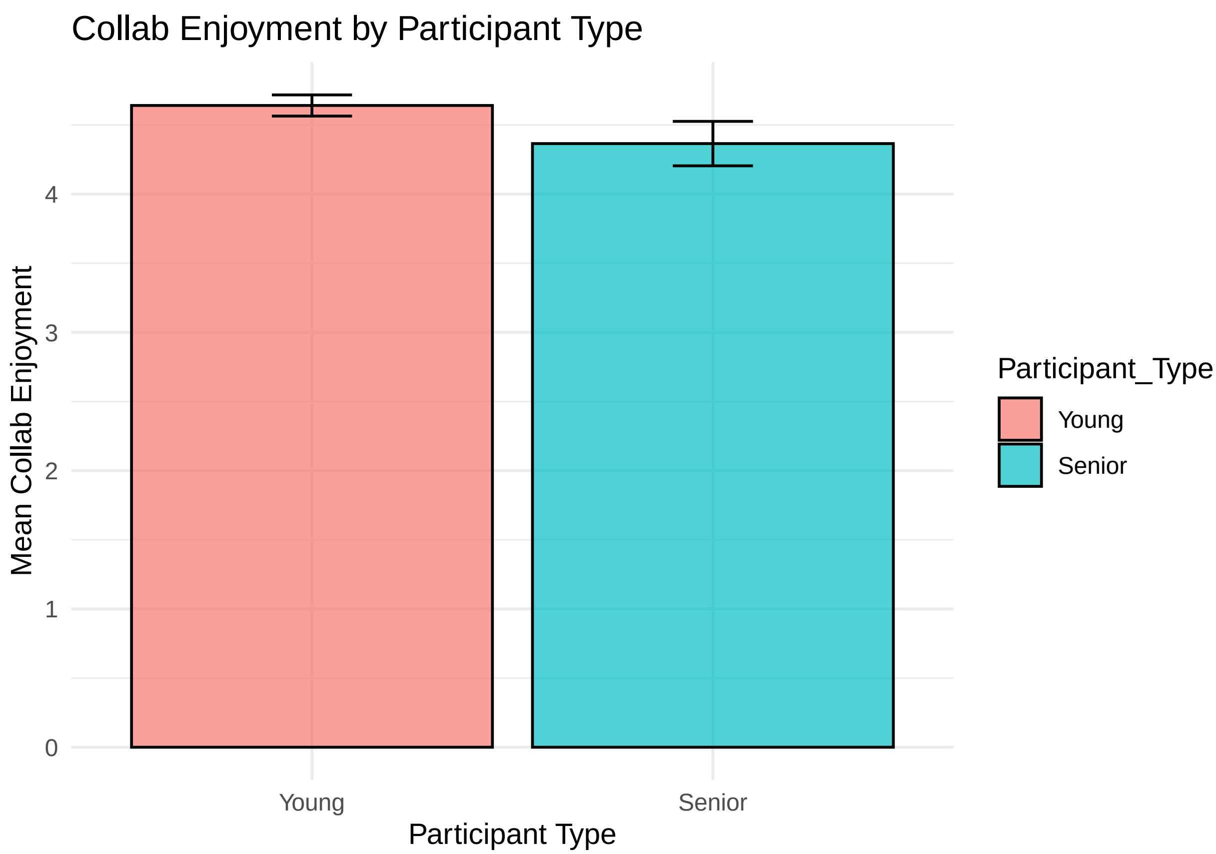
```r
cleaned_data$Participant_Type <- factor(cleaned_data$Participant_Type, levels = c("Young", "Senior"))

# Compute means and standard errors by Survey Type
summary_stats <- cleaned_data %>%
  group_by(survey_type) %>%
  summarise(
    mean_collab_enjoyment = mean(collab_enjoyment, na.rm = TRUE),
    se_collab_enjoyment = sd(collab_enjoyment, na.rm = TRUE) / sqrt(n()),
    mean_pride_outcome = mean(pride_in_outcome, na.rm = TRUE),
    se_pride_outcome = sd(pride_in_outcome, na.rm = TRUE) / sqrt(n())
  )

# Compute means and standard errors by Participant Type
summary_participant <- cleaned_data %>%
  group_by(Participant_Type) %>%
  summarise(
    mean_collab_enjoyment = mean(collab_enjoyment, na.rm = TRUE),
    se_collab_enjoyment = sd(collab_enjoyment, na.rm = TRUE) / sqrt(n()),
    mean_pride_outcome = mean(pride_in_outcome, na.rm = TRUE),
    se_pride_outcome = sd(pride_in_outcome, na.rm = TRUE) / sqrt(n())
  )

# Bar Plot for Collab Enjoyment by Survey Type
ggplot(summary_stats, aes(x = survey_type, y = mean_collab_enjoyment, fill = survey_type)) +
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_collab_enjoyment - se_collab_enjoyment,
                    ymax = mean_collab_enjoyment + se_collab_enjoyment),
                width = 0.2) +
  labs(title = "Collab Enjoyment by Survey Type",
       x = "Survey Type", y = "Mean Collab Enjoyment") +
  theme_minimal()
```



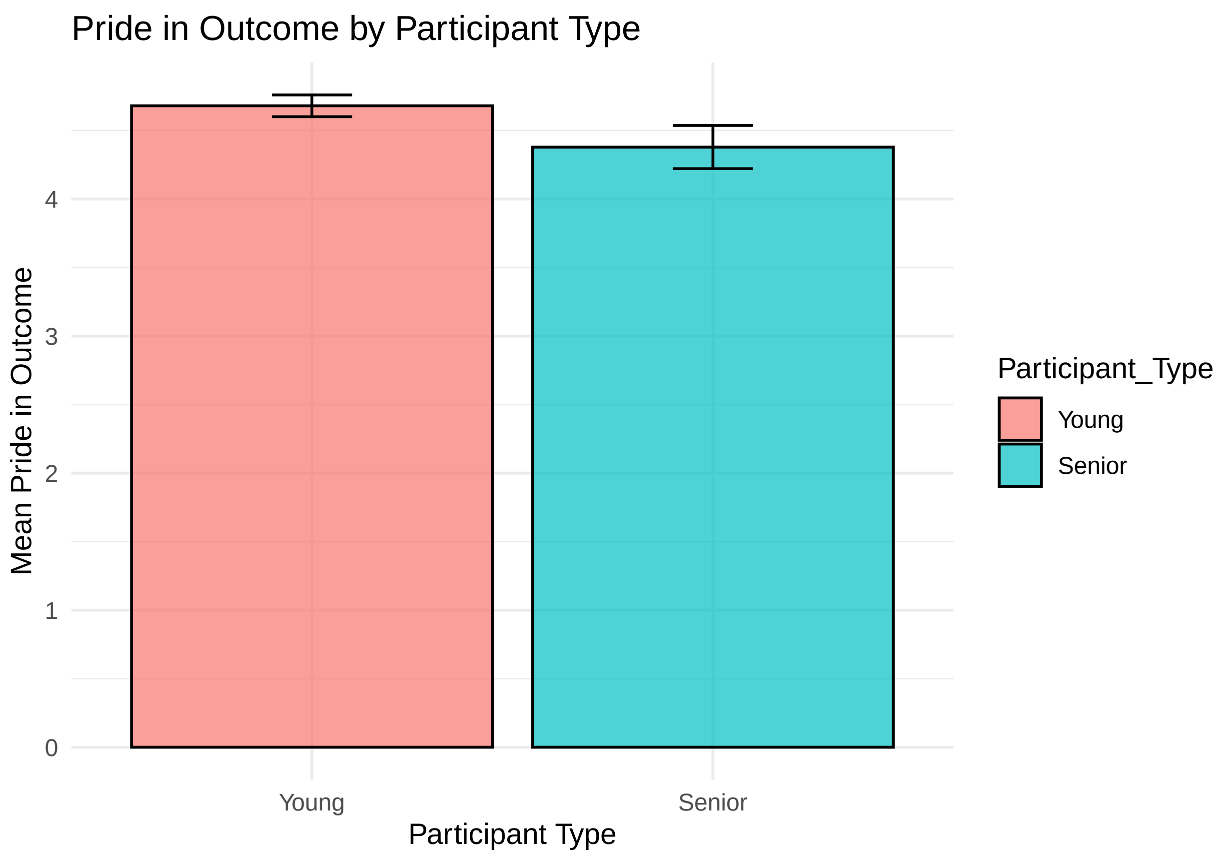

```r
# Bar Plot for Pride in Outcome by Survey Type
ggplot(summary_stats, aes(x = survey_type, y = mean_pride_outcome, fill = survey_type)) +
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_pride_outcome - se_pride_outcome,
                    ymax = mean_pride_outcome + se_pride_outcome),
                width = 0.2) +
  labs(title = "Pride in Outcome by Survey Type",
       x = "Survey Type", y = "Mean Pride in Outcome") +
  theme_minimal()
```




```
# Bar Plot for Collab Enjoyment by Participant Type
ggplot(summary_participant, aes(x = Participant_Type, y = mean_collab_enjoyment, fill = Participant_Type
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_collab_enjoyment - se_collab_enjoyment,
                    ymax = mean_collab_enjoyment + se_collab_enjoyment),
                width = 0.2) +
  labs(title = "Collab Enjoyment by Participant Type",
       x = "Participant Type", y = "Mean Collab Enjoyment") +
  theme_minimal()
```




```r
# Bar Plot for Pride in Outcome by Participant Type
ggplot(summary_participant, aes(x = Participant_Type, y = mean_pride_outcome, fill = Participant_Type))
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_pride_outcome - se_pride_outcome,
                    ymax = mean_pride_outcome + se_pride_outcome),
                width = 0.2) +
  labs(title = "Pride in Outcome by Participant Type",
       x = "Participant Type", y = "Mean Pride in Outcome") +
  theme_minimal()
```


Pride in Outcome by Participant Type
Mean Pride in Outcome
0
1
2
3
4
Young
Senior
Participant Type
Participant_Type
Young
Senior